www.tntextbooks.in

Searching in a one dimensional array:

Searching is a process of finding a particular value present in a given set of numbers. The linear search or sequential search compares each element of the list with the value that has to be searched until all the elements in the array have been traversed and compared.

Program for Linear Search

```
#include <iostream>
using namespace std;
int Search(int arr[], int size, int value)
{
      for (int i=0; i<size; i++)
      {
            if (arr[i] == value)
                  return i; // return index value
      }
      return -1;
}
int main()
{
          int num[10], val, id;
          for (int i=0; i<10; i++)
          {
          cout<< "\n Enter value " << i+1 <<"= ";
          cin>>num[i];
          }
          cout<< "\n Enter a value to be searched: ";
          cin>>val;
          id=Search(num,10,val);
          if(id==-1)
          cout<< "\n Given value is not found in the array..";
          else
          cout<< "\n The value is found at the position" << id+1;
          return 0;
}
```

The above program reads an array and prompts for the values to be searched. It calls Search( ) function which receives array, size and value to be searched as parameters. If the value is found, then it returns the array index to the called statement; otherwise, it returns -1.